

Student 36 **SANGEPU VENKATA PRAVEEN** [192373013RD](#)

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Q1. Smart Document Editor using String, StringBuilder and StringBuffer

CODE

```
import java.util.Scanner;
public class SmartDocumentEditor {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.println("Enter a paragraph:");
        String paragraph = sc.nextLine();
        StringBuffer editHistory = new StringBuffer();
        editHistory.append("Original Document:\n").append(paragraph).append("\n\n");
        int charCount = paragraph.replace(" ", "").length();
        String[] words = paragraph.trim().split("\\s+");
        int wordCount = words.length;
        String[] sentences = paragraph.split("[.!?]+");
        int sentenceCount = 0;
        for (String s : sentences) {
            if (!s.trim().isEmpty()) {
                sentenceCount++;
            }
        }
        editHistory.append("Character Count: ").append(charCount).append("\n");
        editHistory.append("Word Count: ").append(wordCount).append("\n");
        editHistory.append("Sentence Count: ").append(sentenceCount).append("\n\n");
        System.out.println("Enter the word to replace:");
        String oldWord = sc.nextLine();
        System.out.println("Enter the new word:");
        String newWord = sc.nextLine();
        String replacedParagraph = paragraph.replaceAll("\\b" + oldWord + "\\b", newWord);
        editHistory.append("After Replacing ' " + oldWord + "' with ' " + newWord + "':\n");
        editHistory.append(replacedParagraph).append("\n\n");
        String[] sentencesToReverse = replacedParagraph.split("(?<=[.!?])\\s*");
        StringBuilder finalDocument = new StringBuilder();
        editHistory.append("Reversed Sentences:\n");
        for (String sentence : sentencesToReverse) {
            if (sentence.trim().isEmpty()) continue;
            String punctuation = "";
            String sentenceText = sentence.trim();

            if (sentenceText.length() > 0) {
                char lastChar = sentenceText.charAt(sentenceText.length() - 1);
                if (lastChar == '.' || lastChar == '!' || lastChar == '?') {
                    punctuation = String.valueOf(lastChar);
                    sentenceText = sentenceText.substring(0, sentenceText.length() - 1);
                }
            }
            StringBuilder reversed = new StringBuilder(sentenceText);
            reversed.reverse();
            String reversedSentence = reversed.toString() + punctuation;
            editHistory.append(reversedSentence).append("\n");
            finalDocument.append(reversedSentence).append(" ");
        }
        editHistory.append("\nFinal Document Constructed.\n");
        System.out.println("\n----- ORIGINAL DOCUMENT -----");
        System.out.println(paragraph);

        System.out.println("\n----- EDITING HISTORY -----");
        System.out.println(editHistory.toString());

        System.out.println("----- FINAL DOCUMENT -----");
        System.out.println(finalDocument.toString().trim());

        sc.close();
    }
}
```

Report

```
}  
}
```

INPUT

I love Java programming. Java is powerful! Do you like Java?
Java
Python

OUTPUT

(no output)

Q2. Hospital Token Management System using List Interface

CODE

```
import java.util.*;

class Patient {
    int id;
    String name;
    String department;
    Patient(int id, String name, String department) {
        this.id = id;
        this.name = name;
        this.department = department;
    }
    @Override
    public String toString() {
        return id + " " + name + " " + department;
    }
}

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        List<Patient> patientList = new ArrayList<>();
        while (sc.hasNextInt()) {
            int choice = sc.nextInt();
            if (choice == 0) break;
            switch (choice) {
                case 1:
                    int id = sc.nextInt();
                    String name = sc.next();
                    String dept = sc.next();
                    patientList.add(new Patient(id, name, dept));
                    break;
                case 2:
                    for (Patient p : patientList) {
                        System.out.println(p);
                    }
                    break;
                case 3:
                    int updateId = sc.nextInt();
                    String newName = sc.next();
                    String newDept = sc.next();
                    for (Patient p : patientList) {
                        if (p.id == updateId) {
                            p.name = newName;
                            p.department = newDept;
                            break;
                        }
                    }
                    break;
                case 4:
                    int removeId = sc.nextInt();
                    patientList.removeIf(p -> p.id == removeId);
                    break;
                case 5:
                    String searchDept = sc.next();
                    for (Patient p : patientList) {
                        if (p.department.equalsIgnoreCase(searchDept)) {
                            System.out.println(p);
                        }
                    }
                    break;
                case 6:
                    if (!patientList.isEmpty()) {
```

Report

```
                System.out.println(patientList.get(0));
                System.out.println(patientList.get(patientList.size() - 1));
            }
            break;
        default:
            break;
    }
}
sc.close();
}
```

INPUT

```
1
101
John
Cardiology
1
102
Rahul
Neurology
1
103
Meera
Cardiology
2
6
Cardiology
3
102
Rahul Kumar
Oncology
4
101
5
7
0
```

OUTPUT

(no output)

Q3. Airport Boarding Queue Management using Queue Interface

CODE

```
import java.util.*;

class Passenger implements Comparable<Passenger> {
    String name;
    boolean isVIP;
    public Passenger(String name, boolean isVIP) {
        this.name = name;
        this.isVIP = isVIP;
    }
    @Override
    public int compareTo(Passenger other) {
        if (this.isVIP && !other.isVIP) {
            return -1;
        } else if (!this.isVIP && other.isVIP) {
            return 1;
        }
        return 0;
    }
    @Override
    public String toString() {
        return name + (isVIP ? " (VIP)" : " (Regular)");
    }
}

public class AirportQueueManagement {
    public static void main(String[] args) {
        Queue<Passenger> normalQueue = new LinkedList<>();
        PriorityQueue<Passenger> queue = new PriorityQueue<>();
        String[] inputs = {"John (Regular)", "Alice (VIP)", "Bob (Regular)", "David (VIP)", "Emma (Regular)"};

        for (String input : inputs) {
            String name = input.split(" \\(")[0];
            boolean isVIP = input.contains("VIP");
            Passenger p = new Passenger(name, isVIP);
            if (isVIP)
                vipQueue.add(p);
            else
                normalQueue.add(p);
        }
        System.out.println("Current Regular Queue: " + normalQueue);
        System.out.println("Current VIP Queue: " + vipQueue);
        if (!vipQueue.isEmpty())
            System.out.println("Next Passenger: " + vipQueue.peek());
        else if (!normalQueue.isEmpty())
            System.out.println("Next Passenger: " + normalQueue.peek());
        while (!vipQueue.isEmpty()) {
            Passenger boarded = vipQueue.poll();
            System.out.println("Boarded: " + boarded);
            System.out.println("Remaining VIP Queue: " + vipQueue);
        }
        while (!normalQueue.isEmpty()) {
            Passenger boarded = normalQueue.poll();
            System.out.println("Boarded: " + boarded);
            System.out.println("Remaining Regular Queue: " + normalQueue);
        }
    }
}
```

INPUT

```
5
John false
Alice true
Bob false
```

Report

David true
Emma false

OUTPUT

(no output)

Q4. Digital Certificate Verification using Set Interface

CODE

```
import java.util.*;

public class CertificateVerification {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        Set<String> certificateSet = new LinkedHashSet<>();
        int n = sc.nextInt();
        sc.nextLine();
        for (int i = 0; i < n; i++) {
            String certId = sc.nextLine();
            certificateSet.add(certId);
        }
        System.out.println("Certificates in insertion order:");
        for (String id : certificateSet) {
            System.out.println(id);
        }
        System.out.println("Certificates in sorted order:");
        Set<String> sortedSet = new TreeSet<>(certificateSet);
        for (String id : sortedSet) {
            System.out.println(id);
        }
        System.out.println("Total unique certificates: " + certificateSet.size());
        System.out.println("Enter certificate ID to verify:");
        String verifyId = sc.nextLine();
        if (certificateSet.contains(verifyId)) {
            System.out.println("Certificate does not exist.");
        }
        System.out.println("Enter revoked certificate ID:");
        String revokedId = sc.nextLine();
        if (certificateSet.remove(revokedId)) {
            System.out.println("Certificate " + revokedId + " removed.");
        } else {
            System.out.println("Certificate not found.");
        }
        sc.close();
    }
}
```

INPUT

5
C101
C102
C103
C102
C104
C103
C102

OUTPUT

(no output)

Q5. Smart Inventory Management using Map Interface

CODE

```
import java.util.*;

class Product {
    String id;
    String name;
    String category;
    int quantity;
    double price;
    boolean isDiscontinued;

    public Product(String id, String name, String category, int quantity, double price, boolean isDiscontinued) {
        this.id = id;
        this.name = name;
        this.category = category;
        this.quantity = quantity;
        this.price = price;
        this.isDiscontinued = isDiscontinued;
    }

    @Override
    public String toString() {
        return "ID: " + id + ", Name: " + name + ", Category: " + category +
            ", Quantity: " + quantity + ", Price: " + price;
    }
}

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        Map<String, Product> inventory = new HashMap<>();
        if (!sc.hasNextInt()) return;
        int n = sc.nextInt();
        for (int i = 0; i < n; i++) {
            String id = sc.next();
            sc.nextLine();
            String name = sc.nextLine();
            String category = sc.nextLine();
            int qty = sc.nextInt();
            double price = sc.nextDouble();
            boolean discontinued = sc.nextBoolean();
            Product p = new Product(id, name, category, qty, price, discontinued);
            if (!discontinued) {
                inventory.put(id, p);
            }
        }
        if (sc.hasNext()) {
            String queryId = sc.next();
            if (sc.hasNextInt()) {
                int updateQty = sc.nextInt();
                if (inventory.containsKey(queryId)) {
                    Product p = inventory.get(queryId);
                    p.quantity = updateQty;
                }
            }
        }
        for (Product p : inventory.values()) {
            System.out.println(p);
        }

        sc.close();
    }
}
```

INPUT

```
3
P101
Laptop
```



```
Electronics
10
50000
false
P102
Mouse
Accessories
25
500
false
P103
Keyboard
Accessories
15
1500
true
P102
30
```

OUTPUT

(no output)